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CBDC Settlement: Implications for Money Creation and Maturity Transformation

*Francesca Carapella*¹

Abstract

This paper analyzes the implications of different models of intermediation, clearing, and settlement of central bank digital currency (CBDC) for the ability of the financial system to create money and, through it, to possibly affect financial stability. Only two models have the potential to limit the private sectors' ability to create money: a model where the central bank directly opens accounts with the public and a model where the central bank allows intermediaries to perform only custodial activities in CBDC. Under any other intermediation model, money can be created inside the financial system by the intermediaries providing payment services associated with CBDC holdings. However, the creation of inside money does not necessarily affect financial stability per se; it does so indirectly by facilitating activities linked to maturity transformation.

Keywords: banking, central bank digital currency (CBDC), financial stability, payment systems

JEL Classifications: E42, E51, G20

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I. Introduction

Recent technological advances have sparked a wave of new private sector financial products and services, such as digital assets and payment applications, leading central banks around the globe to explore the potential economic value and risks of issuing a central bank digital currency (CBDC). Rules governing the distribution of CBDC and the provision of financial services linked to CBDC are likely to shape the future of our financial system with respect to its potential and its intrinsic fragility.

This paper describes various models for settlement of CBDC balances and for the integration of banks and other financial institutions in the provision of financial services linked to CBDC. Settlement is an act that discharges obligations of funds or securities transfers between two or more parties in accordance with the terms of the underlying contract.² For example, the seller transfers securities to the buyer and the buyer transfers money to the seller.

The settlement models analyzed in this paper differ by their ability to support two important activities typically performed by financial institutions: money creation and maturity transformation. Key elements of traditional payment systems, such as real time gross settlement and deferred net settlement, play a crucial role in determining the extent to which a CBDC settlement model will allow financial institutions to provide liquidity to the economy via money creation and to conduct maturity transformation—a profitable activity that can, however, undermine financial stability.³

The economic mechanism linking CBDC to maturity transformation works through the execution of CBDC payment orders. Payments can often be executed by saving on the amount to be transferred back and forth among trading partners, for example by aggregating and netting payment requests before settlement. In doing so, the entity performing settlement does not need to hold as much CBDC as required by every individual payment order. Hence, a payment provider would be able to invest the CBDC received from clients placing payment requests but not transferred for settlement. A payment provider would be able to perform maturity transformation.

In our financial system, the mechanism just described is already operating through central bank reserves. As a form of central bank money, reserves carry no credit risk and are thus typically preferred in settlement. However, central bank reserves can be held only by institutions eligible for an account at the central bank, typically banks.⁴ A CBDC would also be central bank money. The introduction of a CBDC might introduce additional agents besides banks that can hold a digital version of central bank money, thus creating other

² See Committee on Payments and Market Structure (2016).

³ While this paper highlights ideas linked to financial stress, it focuses on normal rather than crisis times. Therefore, stress episodes like runs on banks are not specifically discussed.

⁴ In the United States, a list of depository institutions eligible for an account is available through the Federal Reserve Master Account and Services Database.

entities that could execute payment orders, in turn facilitating maturity transformation. With maturity transformation having almost always been at the root of the financial crises of the past 200 years, the design of resilient payment systems for CBDC should account for the emergence of new channels that can amplify financial stress by introducing novel points of fragility within and outside the financial system, as some of these entities are often fintech companies.

The analysis shows that CBDC settlement models relying exclusively on the central bank to handle account activity, such as maintaining and updating balances, preclude financial institutions from creating money and operating with a fractional reserve system for CBDC. Therefore, such settlement models would preclude financial institutions from performing maturity transformation through their CBDC-related activities.⁵ As a result, while financial institutions may still create money and engage in maturity transformation through their reserve balances at the central bank, the introduction of a CBDC would not represent an additional channel by which money creation and maturity transformation would be feasible. On the other hand, the analysis also shows that CBDC settlement models allowing for the provision of deposit and payment services by intermediaries can enable both money creation and maturity transformation through CBDC-related products and services. Hence, such settlement models would allow CBDC to introduce an additional channel through which financial institutions can engage in money creation and maturity transformation, possibly complementary to the channels that are currently active. As a result, the choice of settlement models for any potential CBDC can play a key role in shaping the future of our financial system and have far-reaching consequences for financial stability.

This paper is organized as follows: I first describe the process of money creation by financial institutions; second, we discuss maturity and liquidity transformation and their relationship with fractional reserve banking. Last, we define CBDC and describe various settlement models for CBDC, analyzing how each of them might or might not enable money creation and maturity transformation.

II. Money Creation

Money creation is the process that allows the value of transactions in an economy to exceed the stock of central bank money in a unit of time.⁶ In other words, money is created endogenously in an economy when the economy's ability to pay is larger than its stock of money (that is, cash and reserves at the central bank). How is this feasible? Through the provision of payment services by financial institutions when the payment orders stemming

⁵ This settlement model, however, is inconsistent with the design features outlined in the Federal Reserve Board's CBDC white paper, which states that a potential US CBDC would be intermediated (Fed 2022a).

⁶ This note refers to money in its strictest form of payment instrument or medium of exchange. It should be noted, however, that several financial and nonfinancial institutions can be thought of as de facto engaging in activities that affect the economy's ability to borrow or fund productive projects.

from these services do not necessarily rely on central bank money to be settled. Two types of payment services satisfy this condition.⁷

The first type of payment service allowing for money creation is a circulating liability issued by a bank or payment provider. Such liability may circulate in the economy without ever being returned to the issuer in exchange for central bank money. For example, a check could be endorsed and circulate, to the extent that the payees themselves expect to be able to spend it, or to deposit it at their own bank, or to receive central bank money if they return it to the issuing bank.⁸ The check would be used like cash in this situation. Thus, it would be a form of money issued by private institutions, produced inside the financial system, and known as *inside* money, as opposed to central bank money, also known as *outside* money.⁹

Second, payment orders can be netted across a network of institutions before being executed. The most common example of such netting activity occurs among banks. Bank depositors can use their deposits to settle transactions without the need to withdraw their funds, rather by transferring claims on their funds deposited at the bank. Such funds may not be physically at the bank (in the form of either cash or reserves) as the bank may have invested them in loans or other assets. The claims on depositors' funds at the bank are the bank's obligations to pay, so they are claims on the bank's assets up to the face value of the claims. Consider a check, for example. This check can be (a) deposited at the payee's own bank or (b) returned to the issuing bank in exchange for central bank money (cash or reserves). In the first case, the bank where the check is deposited can return the check to the issuing bank and request payment in central bank money. However, when sufficiently many banks engage in check issuance, they can choose to net each other's checks without the need to transfer central bank money back and forth to each other for each check. Because check payments are settled by netting obligations across banks, there is no need for the issuing bank to hold central bank money in the same amount as the checks it issued—unless the issuing bank is certain that none of its depositors will deposit checks issued by other banks, as those checks would be obligations of those other banks toward the issuing bank. More generally, if sufficiently many banks provide payment services associated with deposits, such banks can multilaterally net each other's obligations: banks' outgoing payments are netted against incoming payments from other banks. As a result, full backing of outgoing payments by central bank money (that is, cash and reserves) is not necessary. Such netting of banks' checks effectively generates money. Each dollar on a check issued by a bank and settled by a process of netting obligations across banks is a dollar of money created by the

⁷ The most basic way in which deposits are used to settle transactions is by allowing depositors to withdraw the money that they deposited at their bank and then use that money to pay. This arrangement requires that sufficient central bank money (that is, cash) be kept at the bank to back one-to-one the value of the transactions for which deposits are paying. In this case, the provision of payment services does not result in money creation.

⁸ This used to happen in the Free Banking Era, when banknotes issued by individual banks circulated even outside the area where the issuing bank's main business was located. Some banknotes were crossing states' boundaries, for instance—see Gorton (1999) and Ales et al. (2008).

⁹ For a definition and discussion of inside and outside money see, among many, Mills (2007), Cavalcanti and Wallace (1999), Marimon, Nicolini, and Teles (2003).

banking system, that is, inside money. Additional examples of such process of money creation are historical clearinghouses, as well as central counterparties.¹⁰

We should note that there is a misperception among some economists about money being created *out of thin air*, also known as the *fountain pen money* view of banking.¹¹ According to this view, banks create money by extending a loan to a borrower, as they open an account for the borrower with funds available to be spent in the same amount as the value of the loan. Because the borrower can make payments using the funds in such account, the fountain pen money view of banking is that banks create money by making loans. However, this is not quite accurate. Banks create money by providing deposit accounts to their borrowers only to the extent that the payment orders made from such accounts are netted against incoming payments from other banks, without the need for central bank money.¹² The bank originating the loan will need central bank money for any outgoing payments it is unable to net. This need for central bank money, or *outside* money, implies that the bank originating the loan will not be creating *inside* money simply by opening an account for the borrower. Hence, the bank will not be creating money *out of thin air*.

III. Maturity and Liquidity Transformation and Fractional Reserve Banking

Financial institutions perform maturity transformation when the maturity of their financing does not match the maturity of their investments, for example by promising their funding providers short-term repayments and investing the borrowed funds into longer-term assets. Financial institutions perform liquidity transformation when they invest in illiquid assets and raise funding by issuing liabilities that are broadly regarded as liquid by financial markets, often owing to the issuer's promise to act to preserve such liquidity. The transformation of claims over long-term or fundamentally illiquid assets into claims that are highly liquid is one of the critical functions that the financial sector provides for the broader economy.¹³

Typically, long-term assets are less liquid than short-term ones, thus implying that maturity transformation is a sufficient condition for liquidity transformation. However, liquidity transformation can occur even when both assets and liabilities are short term. Consider, for example, an entity investing in US Treasuries and issuing liabilities that circulate as means of payment, or promising repayment of its liabilities in central bank money. Both assets and

¹⁰ See, among many, Kroszner (2006), Quinn and Roberds (2008), and Moser (1998). Instead, banks' liabilities that are fully backed by central money are not considered inside money for the purpose of this note, even though the mapping between banks' liabilities and central bank money might become imperfect in stress scenarios.

¹¹ See Faure and Gersbach (2021); McLeay, Radia, and Thomas (2014); Jakab and Kumhof (2015); and Deutsche Bundesbank (2017).

¹² See Rendahl and Freund (2019).

¹³ See Wallace (1996) for a formal analysis of such functions.

liabilities are short term, but the liquidity of its liabilities, which are payment instruments, is higher than that of its assets, which would need to be sold and converted into payment instruments to be used as settlement assets.

The most common example of maturity transformation is the supply of demand deposit contracts by banks, while investing deposited funds in longer-term assets. Banks are not the only institutions that engage in maturity and/or liquidity transformation. Many collective investment vehicles offer a degree of liquidity to investors but invest in assets that are not regarded as equally liquid by financial markets, for example money market funds and insurance companies.¹⁴ Banks can perform maturity and liquidity transformation in two ways, possibly complementary with each other:

- (a) Banks diversify depositors' payments and liquidity needs.¹⁵ In the basic demand deposit contract, a depositor experiencing a liquidity need can withdraw the cash previously deposited at the bank and make a payment using such cash. As only some depositors will experience a liquidity need at any given point in time, a bank's need for liquidity is smaller than its aggregate deposits, allowing the bank to invest part of the funds deposited by end users in long-term and more productive assets, such as loans to firms.

Moreover, if the distribution of liquidity needs is known, and if the deposit base is sufficiently large, how many depositors will experience a liquidity need at a given point in time is known in advance.¹⁶ In this setting, there is no aggregate risk, and depositors with no liquidity needs effectively extend credit to depositors with liquidity needs. This extension of credit is risk free in aggregate for the bank, as long as every depositor is paid by the bank when he actually needs liquidity and no bank run occurs.

- (b) Banks issue liabilities that provide payment services. Specifically, banks issue deposits that can be used to settle transactions, thus creating money as described in the previous section. While money creation per se is separate from maturity transformation, it can facilitate maturity transformation by allowing banks to settle payments through netting their respective obligations. Through such netting, banks can execute and settle transactions by relying more on their network of payment flows and less on central bank money, which is necessary only to settle payments that cannot be netted. The resulting reduction in banks' need to hold central bank money increases banks' ability to invest the funds deposited by customers, which are short-term liabilities, into long-term productive investments.

¹⁴ For money market funds, among many, see Schmidt, Timmermann, and Wermers (2016); Cipriani and La Spada (2021); Sunderam (2015); Greenwood, Hanson, and Stein (2015); and Greenwood, Hanson, and Stein (2018). For insurance companies, see Foley-Fisher et al. (2020).

¹⁵ As described in the seminal work of Diamond and Dybvig (1983) and in Nobel Prize Committee (2022).

¹⁶ In this case, the law of large numbers holds. By the law of large numbers, the average liquidity needs for a depositor is also the aggregate or realized liquidity need at the bank.

Without the provision of payment services associated with deposits (activity b), banks could still perform maturity transformation thanks to their diversification activity (activity a). In this case, however, banks need to pay cash to those depositors who demand to withdraw their funds and make payments. To do so, banks need to invest at least some funds in central bank money, such as cash, and to disburse central bank money whenever depositors request a withdrawal.¹⁷ As a consequence, a bank needs to invest in as much central bank money as the expected liquidity needs of its depositors. Therefore, without the provision of payment services associated with deposits, a bank would not be creating inside money. In this case, the settlement assets used by depositors to pay would merely be different forms of central bank money, be it cash or CBDC if one was issued, and the economy needs as much central bank money as the aggregate value of transactions.

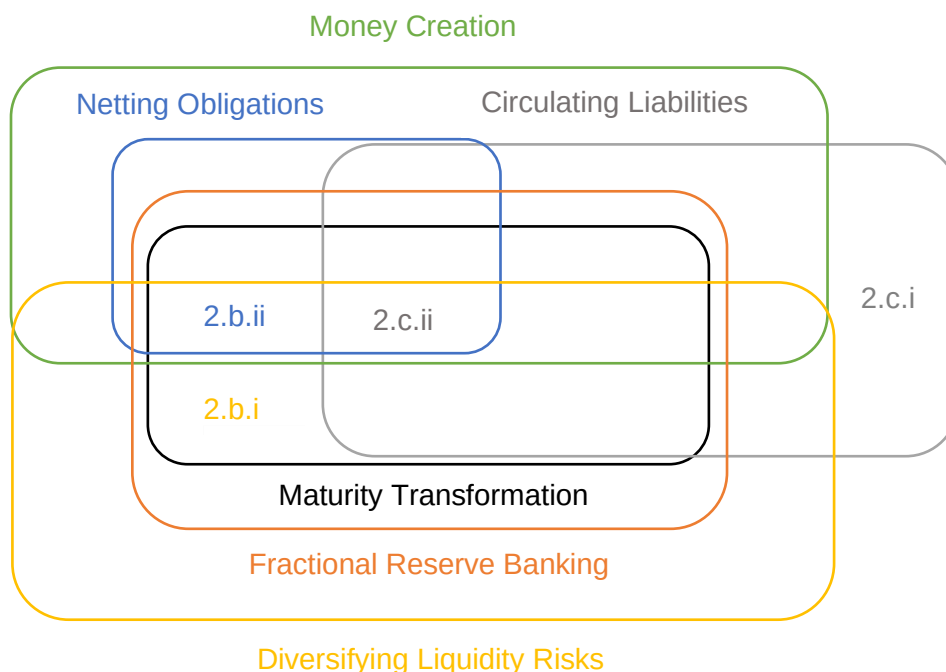
Moreover, without the provision of payment services associated with deposits, a bank might still be fractionalized. A fractional reserve banking system is a system in which banks hold central bank reserves as a fraction of specific types of claims outstanding on its total assets, in particular as a fraction of its demand deposits.¹⁸ Fractionalized banking is linked to maturity transformation, as a bank holding 100% of its deposits in reserves is unable to invest in other long-term assets.¹⁹ Some economists interchangeably discuss maturity transformation and fractional reserve banking. Indeed, a bank whose business model allows for fractional reserves will likely find investing in longer maturities profitable—conditional on a given risk-return trade-off. Here, however, we keep the two definitions separate and note that maturity transformation is a subset of fractional reserve banking (see Figure 1). Indeed, a bank holding a fraction of its deposits in reserves at the central bank, and investing the rest in US Treasuries, would operate with a fractional reserve system despite performing no maturity transformation.²⁰

¹⁷ Banks also hold high-quality liquid assets (HQLAs), but the conversion of HQLAs into reserves to cover net outflows has a similar effect to that of using already held reserves to cover the outflows.

¹⁸ The Federal Reserve Act authorizes the Board to impose reserve requirements on transaction accounts, nonpersonal time deposits, and Eurocurrency liabilities.

¹⁹ In other words, for deposit-taking institutions, a fractionalized system is a necessary condition for them to perform maturity transformation.

²⁰ Such bank, however, could be performing liquidity transformation as the liquidity of its deposits, which provide payment services, would exceed that of its assets, which cannot be used for settlement.

Figure 1: Money Creation, Risk Diversification, and Maturity Transformation

Note: The labels 2.b.i, 2.b.ii, 2.c.i, and 2.c.ii refer to CBDC settlement models described in Section IV.

Source: Author's analysis.

Notice that while a fractional reserve system is a necessary condition for money creation and maturity transformation, money creation is not a necessary condition for maturity transformation. Indeed, money is created when the value of transactions exceeds the stock of central bank money in the economy in a unit of time, which allows banks to hold less reserves than the sum of claims outstanding on those reserves. However, maturity transformation can be performed also by nonbank financial institutions, such as insurance companies, which do not create money in the sense of enabling settlement as described in the previous section.²¹ An essential element of maturity transformation is the different maturity of an institution's assets and liabilities. As argued earlier, banks may choose not to provide payment services—thus not to create inside money—but they will still be able to perform maturity transformation by diversifying the liquidity needs of their depositors, as formalized by Diamond and Dybvig (1983).

Finally, but importantly, while banks have been known to be intrinsically fragile and prone to runs, it should be noted that the main cause for fragility is not money creation per se but maturity and/or liquidity transformation. Indeed, policymakers consider a financial stability concern also the maturity transformation performed by nonbanks despite nonbanks' potentially being less active in the business of money creation than banks. However, money

²¹ See Foley-Fisher, Narajabad, and Verani (2020) and McDonald and Paulson (2015).

creation can facilitate maturity transformation. Figure 1 shows that maturity transformation can be performed by money creation (like banks do) or by diversifying liquidity risks (like banks and also other institutions do), or both, as shown by the intersection between the green and yellow boxes representing, respectively, money creation and diversification of liquidity risks. Figure 1 also represents the ideas described so far: money creation can be performed either by netting obligations (represented by the blue box) or by issuing circulating liabilities (represented by the gray box), and maturity transformation is a subset of fractional reserve banking.

IV. CBDC and Settlement

CBDC is a form of digital money, denominated in the national unit of account, which is a direct liability of the central bank, thus central bank money.²² CBDCs can be designed for use either among financial intermediaries only (that is, wholesale CBDCs), or by the wider economy (that is, retail CBDCs). While the aim of retail CBDCs is to create a digital form of central bank money that can be used by everyone, wholesale CBDC experimentations have so far focused on improving the safety and efficiency of digital interbank transactions, such as securities settlement and cross-border payments.²³ For each CBDC settlement model analyzed in this section, we discuss its potential to be consistent with wholesale or retail CBDC. Importantly, however, the economic mechanisms responsible for money creation and, possibly, maturity transformation are the same both for wholesale and retail CBDC.

With respect to CBDC, the key settlement models can be defined as (a) involving only the activities of the central bank or as (b) intermediated by financial institutions. An intermediated settlement model, in turn, can be characterized by three subcategories, each characterized by a variety of subtleties relating to the role played by financial institutions in the economy. Overall, we analyze five settlement models.

Only the Central Bank Is Involved

In a settlement model where the central bank tracks balances and transfers only at the end-user level, all end users would be allowed to open an account at the central bank, including individuals. Therefore, this settlement model could implement a retail CBDC as well as a wholesale CBDC. On the one hand, if the central bank restricted activities of financial institutions to be retail only, then this settlement model would implement solely a retail CBDC.²⁴ On the other hand, if the central bank offered CBDC balances to be held at accounts

²² Central bank money refers to money that is a liability of the central bank. In the United States, there are currently two types of central bank money: physical currency issued by the Federal Reserve and digital balances held by commercial banks at the Federal Reserve. See Fed (2022b) and Panetta (2022).

²³ For more details, see the projects undertaken by the BIS Innovation Hub (BIS n.d.). Also, see CBDC Tracker (n.d.).

²⁴ Central banks are exploring safeguards that could be built into any CBDC to address financial stability risks, such as access criteria for permitted users and limits on individuals' CBDC holdings or transactions, which could

at the central bank only to financial institutions, then this settlement model would implement solely a wholesale CBDC.

With banks or other payment service providers not involved in any CBDC payment-related activity, there would be no inside money creation stemming from CBDC issuance. In fact, no netting of CBDC payment orders would take place across a network of intermediaries, nor would intermediaries' liabilities backed by CBDC circulate as a medium of exchange. Moreover, if intermediaries were ineligible to hold CBDC, no maturity transformation stemming from CBDC issuance would take place. Any maturity transformation activity in which intermediaries engaged would stem from their holdings of other forms of central bank money. Therefore, there would be no place for this settlement model in Figure 1, while in Figure 2, this is defined as settlement model 1.

However, it is worth noting that even with the central bank tracking CBDC balances at the user level, and with intermediaries acting as end users, it is possible that intermediaries might develop forms of recordkeeping and settlement of CBDC balances that operate separately from the central bank's record. In doing so, banks might offer payment services on such parallel recordkeeping platforms, submitting to the central bank settlement system only the net transfer resulting from all the transactions recorded on their own platforms. Thus, the possibility to create such alternative recordkeeping platforms would transform this settlement model into an intermediated one, where payment services could be provided, possibly money could be created, and maturity transformation could be performed.

Intermediated

Intermediaries act as custodians.²⁵ In Figure 2, this is defined as settlement model 2.a, where intermediaries maintain end-user accounts or wallets and track end-user balances. The central bank processes CBDC balances only at the intermediary level. According to this settlement model, intermediaries would be effectively custodians and distributors of CBDC, thus implementing a retail CBDC. With intermediaries simply maintaining end-user accounts or wallets without proprietary CBDC holdings, there would be no money creation stemming from CBDC issuance, as their role in the CBDC ecosystem would purely be custodial and recordkeeping. As with the first settlement model, there would be no place for this settlement model in Figure 1, as the implications for money creation and maturity transformation from both models are the same. Notice that this settlement model would differ from the first one with banks acting as end users, as no proprietary holdings are considered here when banks simply act as custodians.

be applied varyingly for different CBDC account holders and perhaps contingent on stress events. See BIS (2021).

²⁵ The entities providing intermediation need not be financial institutions but may be any institution allowed by policymakers to perform the functions specified in each settlement model. Hence, payment and wallet providers, even telecommunications and social media companies that have access to (retail) CBDC might be able to intermediate the transfer or safekeeping of CBDC.

Intermediaries act also as end users and offer on-balance-sheet digital liabilities. When intermediaries participate in the CBDC ecosystem as end users, they are allowed proprietary holdings of CBDC beyond holding CBDC balances for their clients. Hence, intermediaries might choose to use CBDC as a substitute for reserves to settle payments and to meet regulatory requirements, if eligible to do so. Under this settlement model, intermediaries might choose to offer deposit contracts in CBDC and to offer payment services associated with the deposit contracts, such as digital wallets or payment application program interfaces (APIs).²⁶

This settlement model would implement both a retail and a wholesale CBDC. Retail CBDC would be distributed to the general public by intermediaries as in the previous settlement model (2.a) and could be held by intermediaries themselves for proprietary trading. Wholesale CBDC would be limited to the settlement of transfers across intermediaries originating either from proprietary trading or from clients' payments associated with deposit contracts.

This settlement model could greatly affect the ability of intermediaries to create inside money and to perform maturity transformation.

Consider first the case where the deposit contracts offered by intermediaries are not associated with any payment service, while they might offer remuneration, safekeeping, or other bundles of products. In Figures 1 and 2, this is referred to as settlement model 2.b.i. With the lack of payment services, no money creation stemming from CBDC issuance would be feasible for intermediaries. However, intermediaries might still engage in diversification of users' liquidity risks, if they are able to attract sufficiently many depositors, and possibly in maturity transformation. Therefore, this settlement model would be located in the lower half of Figure 1, possibly within the black box if maturity transformation is performed. In particular, intermediaries could perform maturity transformation through their offering of both deposit contracts in CBDC—if they are not subject to a 100% CBDC reserve requirement—and deposit contracts in cash or reserves, as is currently the case. With respect to the current environment, however, this settlement model would allow for an additional channel by which maturity transformation can be performed, working through CBDC deposits not 100% backed by CBDC.

Consider now the case where intermediaries offer payment services associated with deposit contracts. In Figures 1 and 2, this is referred to as settlement model 2.b.ii. Besides diversification of liquidity risks, intermediaries could also engage in money creation. They could do so by simply netting payment orders among themselves before settlement takes place in CBDC, much like banks do now by netting each other's obligations before settlement takes place in central bank reserves. By virtue of money creation, intermediaries then would

²⁶ These liabilities could be redeemable in CBDC, depending on the nature of the deposit contract written between the intermediaries and depositors. If they were redeemable in CBDC, then settlement models 2.b.i and 2.b.ii, described later in this section, would be similar to a fixed-term deposit, which could be anywhere from overnight to a longer term, and settlement model to a payment account such as a demand deposit account, respectively.

not need to hold a dollar of CBDC for each payment of a dollar, which would allow them to be fractionalized and engage in maturity transformation. Therefore, within this settlement model, intermediaries could engage in maturity transformation via two channels: diversification of liquidity risks, as in the previous model, and money creation. This settlement model would be located at the center-left of Figure 1, as both money creation and risk diversification could enable maturity transformation, but no liabilities are circulating as media of exchange.

Intermediaries offer circulating on-balance-sheet digital liabilities. If intermediaries offer on-balance-sheet liabilities to end users, these can circulate and be readily transferrable much like cash. Hence, the end user's legal claim is not directly on the central bank but rather on the assets of the intermediaries. The intermediaries' digital liabilities could then be 100% backed by central bank money.²⁷ For example, this settlement model would allow for the issuance of stablecoins fully backed by central bank money. As the digital liability circulates, its transfers would occur on the rails on which it is designed to operate, possibly independent of the payment and settlement rails for the underlying CBDC. Redemption of the digital liability would still require wholesale or retail CBDC settlement rails as, ultimately, CBDC would need to be paid to the holder of the liability to extinguish all obligations of the issuer.²⁸ Similarly, issuance of the digital liability would occur on wholesale or retail CBDC settlement rails. Wholesale or retail CBDC rails would be used depending on the type of user requesting issuance or redemption. Alternatively, the intermediaries' digital liabilities could be backed by less than 100% in central bank money, as fractionalized banking.²⁹ As described in Section III, a fractional reserve system allows for settlement of the liabilities issued by intermediaries without the need to transfer the same amount of central bank money for each dollar of an intermediary's liability. As long as the intermediaries' liabilities circulate, no transfer of CBDC occurs because the transfer of the liabilities themselves can extinguish a payment obligation. For example, this settlement model would allow for the issuance of tokenized deposits, which could be used to make payments and could be settled on rails specifically designed for this purpose, independently of the CBDC settlement rails.³⁰ Obligations stemming from the transfer of such tokenized deposits when

²⁷ Some have described versions of this model as synthetic CBDC (s-CBDC). See Auer and Böhme (2020) and Adrian and Mancini Griffoli (2019).

²⁸ It is worth highlighting that intermediaries' ability to offer on-balance-sheet liabilities can be implemented also with other forms of central bank money, such as reserves held at the central bank. Indeed, while no bank liabilities currently circulate as media of exchange, they used to do so in the Free Banking Era (see Gorton [1999] and Ales et al. [2008]). It is conceivable that tokens might circulate in the future, perhaps simply in the form of tokenized deposits. If CBDC were introduced, however, and a broader set of entities than depository institutions were eligible to hold it, then such entities may also issue circulating on-balance-sheet liabilities and settle their respective net obligations in central bank money, as banks can do today with reserve balances held in their accounts at the central bank.

²⁹ In the current payment and settlement systems, where CBDC is not present yet, this model already exists through deposits issued by commercial banks.

³⁰ This paper refers to tokenized deposits as the digital representation of existing liabilities—demand deposit claims—that an intermediary has on its balance sheet. Tokenized deposits are obligations of the intermediary and, besides being used for the internal representation of credits and debits of accounts at the intermediary, can circulate more broadly across intermediaries and users if their holders choose to do so.

returned to the issuer would be settled on either wholesale or retail CBDC rails, depending on the parties involved.³¹

The circulating digital liabilities offered by intermediaries could then be:

- (a) 100% backed by central bank money. In Figures 1 and 2, this is referred to as settlement model 2.c.i. In this case, no money creation would be feasible, despite the intermediaries' liabilities could circulate as media of exchange. Moreover, the 100% reserve requirement for the circulating liability would prevent the intermediary from performing diversification of liquidity risks and maturity transformation through the issuance of this specific liability. Hence, this settlement model would be located outside of both the green and yellow boxes in Figure 1. Notice that this settlement model can be complementary to settlement model 2.b.i, as (noncirculating) CBDC deposits exempt from the 100% reserve requirement might also be issued along with fully backed circulating digital liabilities. In this case, maturity transformation would be feasible but only through the intermediary's activities not subject to the 100% backing by central bank money.
- (b) Backed by less than 100% in central bank money, as fractionalized banking. In Figures 1 and 2, this is referred to as settlement model 2.c.ii. This case is the complement of settlement model 2.b.ii, as the liabilities of an intermediary could also circulate as media of exchange besides being netted in settlement. Therefore, money creation could arise through an additional channel with respect to settlement model 2.b.ii: circulation of liabilities that are not fully backed by central bank money. This settlement model would be located right at the center of Figure 1, as all activities depicted in Figure 1 would be feasible under this settlement model.

Figure 2 summarizes the activities that are and are not feasible for each settlement model. In the first settlement model, where only the central bank is involved, none of the activities described in the previous sections can be performed, as intermediaries are not actively participating in the CBDC ecosystem. The same conclusion applies to the second settlement model, when intermediaries act as custodians (2.a). However, if intermediaries act as end users without offering payment services, as in settlement model 2.b.i, then diversification of liquidity shocks, fractional reserve banking, and maturity transformation are all feasible. This is a world similar to the banking model developed in Diamond and Dybvig (1983). If intermediaries act as end users and offer payment services, as in settlement model 2.b.ii, then money creation via netting obligations is also feasible. Finally, if intermediaries offer circulating on-balance-sheet digital liabilities that are 100% backed by central bank money, as in settlement model 2.c.i, no money creation occurs owing to the full backing of liabilities by central bank money. On the other hand, if the circulating on-balance-sheet digital liabilities are backed by less than 100% central bank money, as in settlement model 2.c.ii,

³¹ When returned to the issuer, tokenized deposits would be settled with the same process currently used to settle payments from deposit accounts—which are either multilaterally netted intraday, thus requiring only the net obligation to be honored in central bank money between the banks involved, or on a real-time-gross-settlement (RTGS) system.

then money creation becomes feasible both via circulation of such liabilities and by netting obligations stemming from the transfer of such liabilities.

Figure 2: Activities That Are Feasible (✓) and Not Feasible (✗) for Each Settlement Model

CBDC Model	Diversifying Shocks	Fractional Reserve Banking	Maturity Transformation	Netting Obligations	Circulating Liabilities	
				Money Creation		No Money Creation
1	✗	✗	✗	✗	✗	✗
2.a	✗	✗	✗	✗	✗	✗
2.b.i	✓	✓	✓	✗	✗	✗
2.b.ii	✓	✓	✓	✓	✗	✗
2.c.i	✗	✗	✗	✗	✗	✓
2.c.ii	✓	✓	✓	✓	✓	✗

Source: Author's analysis.

V. Conclusion

The links among money, whether issued by the central bank or produced by the financial system, maturity transformation, and payment systems are crucial elements for the design of a CBDC settlement model. Some CBDC settlement models do not allow for money creation nor maturity transformation and are, therefore, the least concerning from a financial stability perspective. However, these models require that each dollar spent be settled by transferring central bank money such as CBDC. Hence the velocity of a given stock of CBDC would need to be sufficiently high to support the same value of transactions supported by other settlement models that allow for money creation.³² Settlement models that allow for money creation but not for maturity transformation are not necessarily threatening the stability of the financial system, while they retain advantages in providing some degree of flexibility in the supply of liquidity to the economy given a preexisting stock of central bank money. In fact, setting aside trust issues such as custodial and operational risks as well as fraud, money creation by financial intermediaries does not pose financial stability threats. However, money creation is rarely present without maturity transformation owing to the profitability of the latter, which is made feasible by the former. Settlement models that allow

³² In other words, models not allowing for money creation would be liquidity intensive, like RTGS and prefunded payments systems.

for maturity transformation are the riskiest from a financial stability perspective as they expose the financial system to potentially destabilizing runs.

VI. References

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